

Bash Script Lab 2

System Administration Alexandria

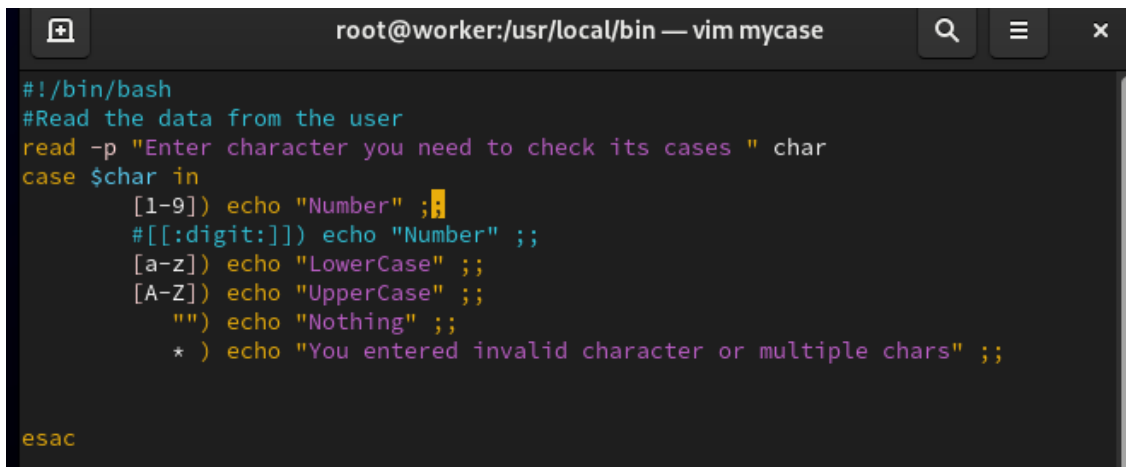
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Q1. Write a script called mycase, using the case utility to checks the type of character entered by a user:

a. Upper Case. b. Lower Case.

c. Number. d. Nothing.

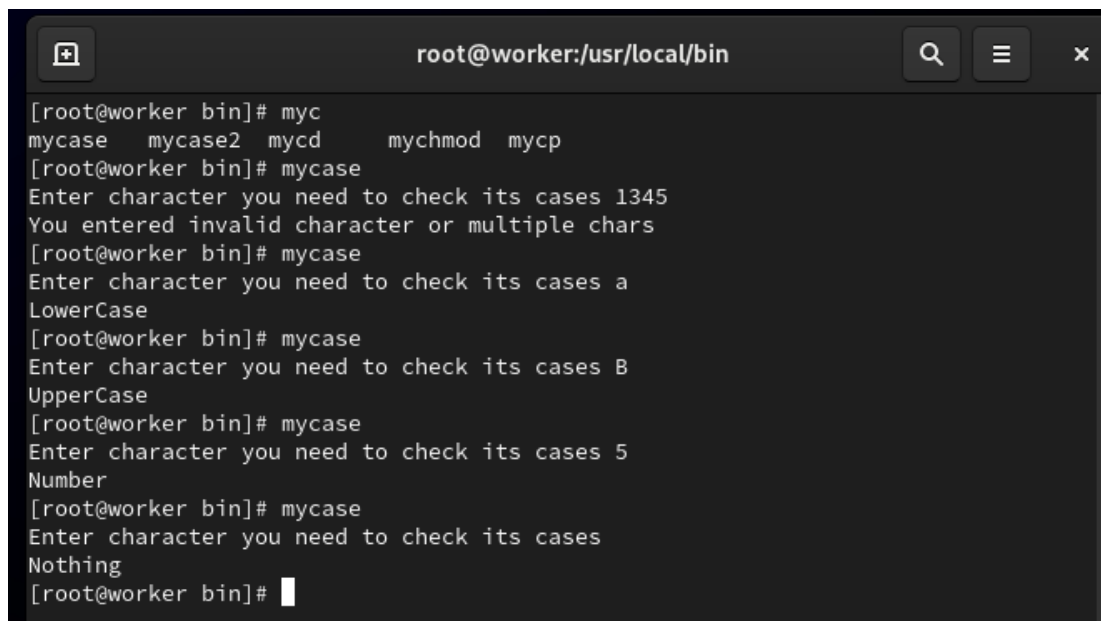
Answer:



```
root@worker:/usr/local/bin — vim mycase

#!/bin/bash
#Read the data from the user
read -p "Enter character you need to check its cases " char
case $char in
    [1-9]) echo "Number" ;;
    #[:digit:]) echo "Number" ;;
    [a-z]) echo "LowerCase" ;;
    [A-Z]) echo "UpperCase" ;;
    "") echo "Nothing" ;;
    * ) echo "You entered invalid character or multiple chars" ;;
esac
```

Verification:



```
root@worker:/usr/local/bin

[root@worker bin]# myc
mycase mycase2 mycd mychmod mycp
[root@worker bin]# mycase
Enter character you need to check its cases 1345
You entered invalid character or multiple chars
[root@worker bin]# mycase
Enter character you need to check its cases a
LowerCase
[root@worker bin]# mycase
Enter character you need to check its cases B
UpperCase
[root@worker bin]# mycase
Enter character you need to check its cases 5
Number
[root@worker bin]# mycase
Enter character you need to check its cases
Nothing
[root@worker bin]#
```

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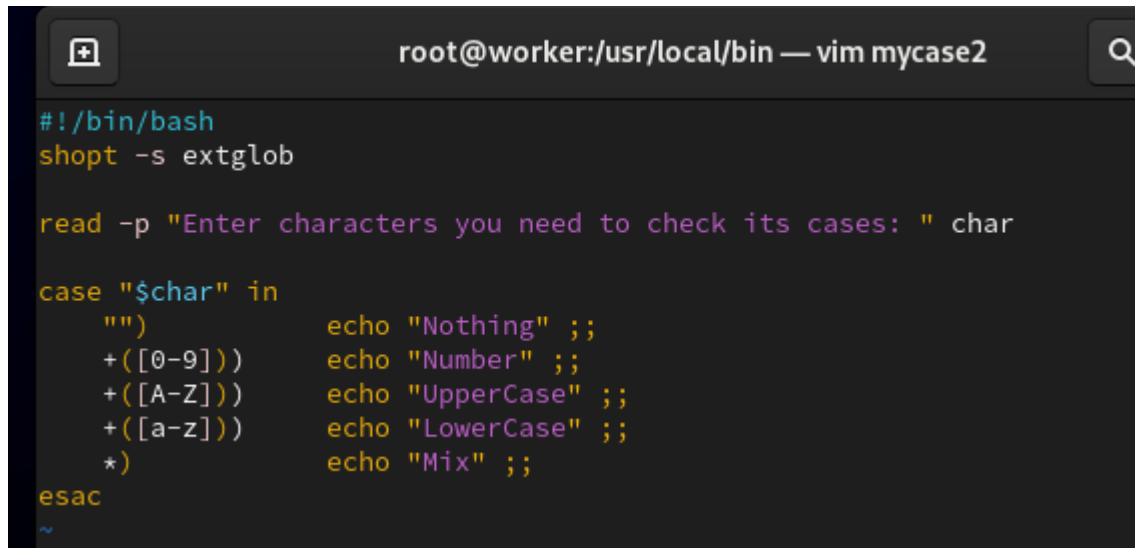
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Q2. Enhanced the previous script, by checking the type of string entered by a user:

a. Upper Cases. b. Lower Cases. c. Numbers.

d. Mix. e. Nothing.

Answer:

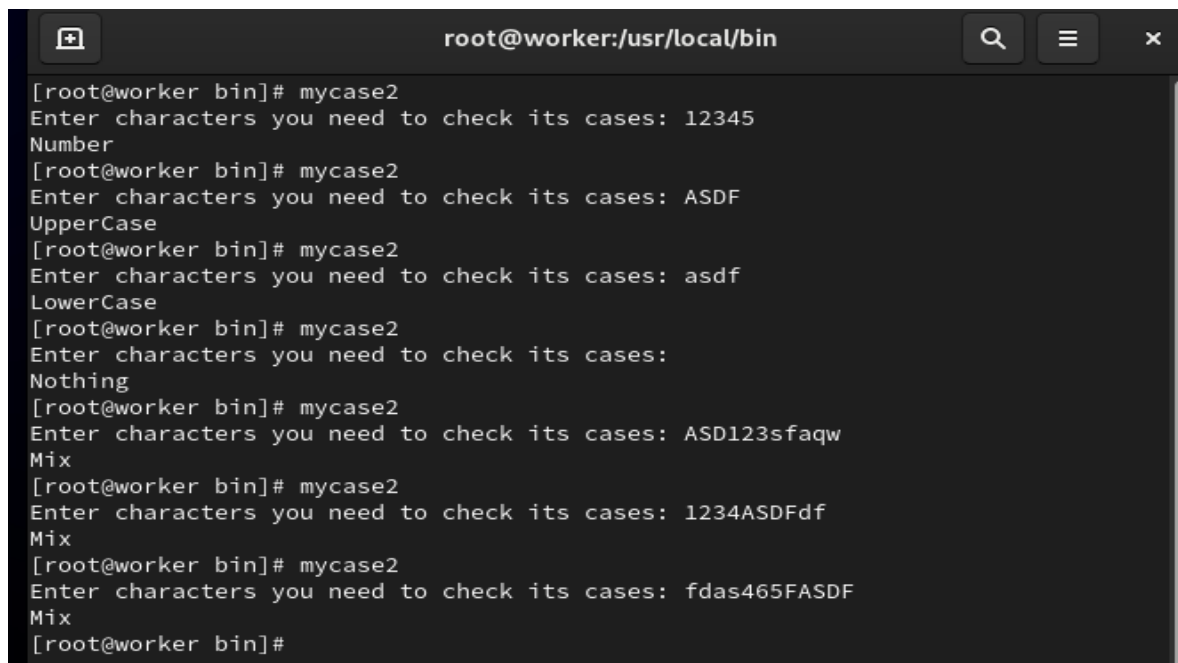


```
#!/bin/bash
shopt -s extglob

read -p "Enter characters you need to check its cases: " char

case "$char" in
    "")          echo "Nothing" ;;
    +([0-9]))    echo "Number" ;;
    +([A-Z]))    echo "UpperCase" ;;
    +([a-z]))    echo "LowerCase" ;;
    *)          echo "Mix" ;;
esac
```

Verification:



```
[root@worker bin]# mycase2
Enter characters you need to check its cases: 12345
Number
[root@worker bin]# mycase2
Enter characters you need to check its cases: ASDF
UpperCase
[root@worker bin]# mycase2
Enter characters you need to check its cases: asdf
LowerCase
[root@worker bin]# mycase2
Enter characters you need to check its cases:
Nothing
[root@worker bin]# mycase2
Enter characters you need to check its cases: ASD123sfaqw
Mix
[root@worker bin]# mycase2
Enter characters you need to check its cases: 1234ASDFdf
Mix
[root@worker bin]# mycase2
Enter characters you need to check its cases: fdas465FASDF
Mix
[root@worker bin]#
```

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Q3. Write a script called mychmod using for utility to give execute permission to all files and directories in your home directory.

Answer:

```
root@worker:/usr/local/bin — vim mychmod
#!/bin/bash
find "$HOME" -exec chmod a+x {} \;
```

Verification:

```
root@worker:~
[root@worker ~]# mychmod
[root@worker ~]# ls -l
total 544
-rwxr-xr-x. 1 root root 438 Apr 7 12:06 addscripts.sh
-rwx--x--x. 1 root root 958 Mar 20 14:33 anaconda-ks.cfg
drwxr-xr-x. 4 root root 58 Apr 8 03:04 backup
-rwx--x--x. 1 root root 6190 Apr 7 09:56 bash.xlsx
drwxr-xr-x. 2 root root 16 Apr 7 02:10 Desktop
drwxr-xr-x. 2 root root 6 Mar 30 03:22 Documents
drwxr-xr-x. 2 root root 6 Mar 30 03:22 Downloads
-rwx--x--x. 1 root root 527785 Apr 2 17:01 ExamQuestions.pdf
-rwxr-xr-x. 1 root root 0 Apr 8 01:04 f1.sh
-rwxr-xr-x. 1 root root 0 Apr 8 01:04 f2.sh
-rwxr-xr-x. 1 root root 0 Apr 8 01:04 f3.sh
-rwxr-xr-x. 1 root root 0 Apr 8 01:04 f4.sh
-rwxr-xr-x. 1 root root 0 Apr 8 01:04 f5.sh
-rwxr-xr-x. 1 root root 134 Apr 7 12:09 input.txt
drwxr-xr-x. 2 root root 6 Mar 30 03:22 Music
-rwxr-xr-x. 1 root root 134 Apr 7 12:01 output1
drwxr-xr-x. 2 root root 6 Mar 30 03:22 Pictures
drwxr-xr-x. 2 root root 6 Mar 30 03:22 Public
-rwxr-xr-x. 1 root root 689 Apr 7 12:21 sendmail.sh
drwxr-xr-x. 2 root root 6 Mar 30 03:22 Templates
drwxr-xr-x. 2 root root 6 Mar 30 03:22 Videos
```

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Q4. Write a script called mybackup using for utility to create a backup of only files in your home directory.

Answer:

```
root@worker:~ — vim /usr/local/bin/mybackup

#!/bin/bash

if [ ! -d ~/backup ]
then
    mkdir ~/backup
fi

dir=~ /backup/$(date +%F\-%H%M\-%M'M')
mkdir -p $dir

cp ~/* $dir 2>/dev/nul
```

Verification:

```
root@worker:~

[root@worker ~]# mybackup
[root@worker ~]# ls -R backup/
backup/:
2026-04-08-03H-02M  2026-04-08-03H-04M  2026-04-08-03H-19M

backup/2026-04-08-03H-02M:
addscripts.sh  bash.xlsx      f1.sh  f3.sh  f5.sh  output1
anaconda-ks.cfg ExamQuestions.pdf f2.sh  f4.sh  input.txt  sendmail.sh

backup/2026-04-08-03H-04M:
addscripts.sh  bash.xlsx      f1.sh  f3.sh  f5.sh  output1
anaconda-ks.cfg ExamQuestions.pdf f2.sh  f4.sh  input.txt  sendmail.sh

backup/2026-04-08-03H-19M:
addscripts.sh  bash.xlsx      f1.sh  f3.sh  f5.sh  output1
anaconda-ks.cfg ExamQuestions.pdf f2.sh  f4.sh  input.txt  sendmail.sh
[root@worker ~]#
```